

8. To Detect Variable References, Expand Values, Handle Undefined Variables, Update Tokens, Support Nested Variables, Test Expansion Logic.

To Identify Built-ins, Create Dispatch Table, Execute In-Process Commands, Handle Invalid Commands, Maintain State, Test Dispatch Logic.

```
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

int main() {
    char input[100];

    while (1) {
        printf("myshell> ");
        fgets(input, sizeof(input), stdin);

        input[strcspn(input, "\n")] = '\0';

        if (strcmp(input, "exit") == 0)
            break;

        if (strcmp(input, "pwd") == 0) {
            char *dir = getenv("PWD");

            if (dir)
                printf("%s\n", dir);

            continue;
        }

        if (strncmp(input, "echo $", 6) == 0) {
            char *var = input + 6;
            char *value = getenv(var);

            continue;
        }

        if (strncmp(input, "echo $", 6) == 0) {
            char *var = input + 6;
            char *value = getenv(var);

            if (value)
                printf("%s\n", value);
            else
                printf("Undefined variable\n");

            continue;
        }

        printf("Unknown command: %s\n", input);
    }

    return 0;
}
```

Output:

```
manisha@LAPTOP-5SMR05RJ:~/SK$ nano skills.c
manisha@LAPTOP-5SMR05RJ:~/SK$ gcc skills.c -o skills
manisha@LAPTOP-5SMR05RJ:~/SK$ ./skills
myshell> pwd
/home/manisha
myshell> ^Z
[2]+  Stopped                  ./skills
manisha@LAPTOP-5SMR05RJ:~/SK$ █
```

09. To Change Directories, Validate Paths, Update Working Directory, Handle Errors, Maintain Previous Directory, Test Navigation.

To Identify Built-ins, Create Dispatch Table, Execute In-Process Commands, Handle Invalid Commands, Maintain State, Test Dispatch Logic.

GNU nano 8.7.1

```
#include <stdio.h>
#include <unistd.h>
#include <string.h>
#include <stdlib.h>
#include <limits.h>

int main() {
    char input[200];

    while (1) {
        char cwd[PATH_MAX];

        getcwd(cwd, sizeof(cwd));

        printf("%s$ ", cwd);
        fgets(input, sizeof(input), stdin);

        input[strcspn(input, "\n")] = '\0';

        if (strcmp(input, "exit") == 0)
            break;

        if (strncmp(input, "cd ", 3) == 0) {
            char *path = input + 3;

            if (chdir(path) == -1)
                perror("cd");
            else
                printf("Directory changed successfully.\n");
        }
        else {
            printf("Command: %s\n", input);
        }
    }

    return 0;
}
```

Output:

```
manisha@LAPTOP-5SMR05RJ:~/SK$ nano skills.c
manisha@LAPTOP-5SMR05RJ:~/SK$ gcc skills.c -o skills
manisha@LAPTOP-5SMR05RJ:~/SK$ nano skills.c
manisha@LAPTOP-5SMR05RJ:~/SK$ gcc skills.c -o skills
manisha@LAPTOP-5SMR05RJ:~/SK$ ./skills
/home/manisha$ cd /tmp
Directory changed successfully.
/tmp$ █
```

10. To Retrieve Current Directory, Display Path, Process Exit Requests, Cleanup Resources, Save State, Verify Correct Behavior.

To Parse Export Syntax, Update Environment Variables, Validate Names, Handle Existing Variables, Export to Child Processes, Test Functionality.

```

GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <limits.h>

int main() {
    char input[200];

    while (1) {
        printf("myshell> ");
        fgets(input, sizeof(input), stdin);

        input[strcspn(input, "\n")] = '\0';

        /* pwd */
        if (strcmp(input, "pwd") == 0) {
            char cwd[PATH_MAX];

            if (getcwd(cwd, sizeof(cwd)) != NULL)
                printf("%s\n", cwd);
        }

        /* exit */
        else if (strcmp(input, "exit") == 0) {
            printf("Exiting shell...\n");
            break;
        }

        /* export */
        else if (strncmp(input, "export ", 7) == 0) {
            char *assignment = input + 7;

            /* export */
            else if (strncmp(input, "export ", 7) == 0) {
                char *assignment = input + 7;

                if (putenv(assignment) == 0)
                    printf("Environment variable updated.\n");
                else
                    perror("export");
            }

            else {
                printf("Unknown command\n");
            }
        }

        return 0;
    }
}

```

Output:

```
manisha@LAPTOP-5SMR05RJ:~/SK$ nano skills.c
manisha@LAPTOP-5SMR05RJ:~/SK$ gcc skills.c -o skills
manisha@LAPTOP-5SMR05RJ:~/SK$ ./skills

myshell> export NAME=Manisha

Environment variable updated.

myshell> █
```
